Name: _____

WORKSHEET

Finding derivatives from first principles

The ideals which have always shown before me and filled me with the joy of living are …

12	10	10	2	5	6	9	9	11	6	4	3	13	1

4	5	2	13	8	3	13	7		

Albert Einstein

Rule

The derivative of a function $f(x)$ at x, if it exists, can be found 'from first principles' by

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Questions

Find the derivatives of the following functions from first principles.

1 $3x + 1$

2 $5x - 2$

3 $7x + 4$

4 $x^2 + 6$

5 $x^2 + 2x - 3$

6 $7x^2 - 11x + 4$

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7 $3x^2 + x - 9$

8 x^3

Find the derivatives of the following functions from first principles at the given points.

9 $x^2 + 2x + 5$ at $x = 5$

10 $3x^2 - 6x + 2$ at $x = 1$

11 $2x + 7$ at $x = 3$

12 $4x^2 - 5x + 1$ at $x = 2$

13 x^3 at $x = 3$

Solutions

A $2x$

B 2

D 5

E $14x - 11$

G 11

H $6x + 1$

N $2x + 2$

O 0

R $3x^2$

S 12

T 27

U 7

Y 3

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